

# 第1章 矢量分析基础

## 1. 方向导数与梯度

$$\nabla u = \frac{\partial u}{\partial x} \vec{e}_x + \frac{\partial u}{\partial y} \vec{e}_y + \frac{\partial u}{\partial z} \vec{e}_z$$

$$\vec{r} = \sqrt{x^2 + y^2 + z^2}$$

$$\nabla r = \frac{x}{r} \vec{e}_x + \frac{y}{r} \vec{e}_y + \frac{z}{r} \vec{e}_z$$

向量  
梯度  $\nabla$  值标量 = 标量分别对  $x, y, z$  求偏导.

$$\vec{r} = x\vec{e}_x + y\vec{e}_y + z\vec{e}_z$$

$$\nabla \cdot \vec{r} = \frac{\partial x}{\partial x} + \frac{\partial y}{\partial y} + \frac{\partial z}{\partial z} = 3$$

散度  $\nabla$  矢量 = 各分量分别对  $x, y, z$  求偏导之和  
标量(值)

$$\nabla \times \vec{r} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x & y & z \end{vmatrix} = \vec{0}$$

旋度  $\nabla \times$  矢量 = 矢量

## 2. 通量与散度

$$\nabla \cdot \vec{A} = \text{div } \vec{A} = \lim_{\Delta V \rightarrow 0} \frac{\oint \vec{A} \cdot d\vec{S}}{\Delta V}$$

通量的微分

$$= \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

散度定理(奥高公式)

$$\oint_S \vec{A} \cdot d\vec{S} = \int_V \nabla \cdot \vec{A} dV$$

面积分 体积分

## 3. 环度与旋度

$$\text{rot } \vec{A} = \nabla \times \vec{A} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

斯托克斯定理

$$\oint_L \vec{F} \cdot d\vec{l} = \int_S (\nabla \times \vec{F}) \cdot d\vec{S}$$

环量 旋度的面积分

## 4. 圆柱坐标系 $(\rho, \phi, z)$

$$\nabla u = \frac{\partial u}{\partial \rho} \vec{e}_\rho + \frac{1}{\rho} \frac{\partial u}{\partial \phi} \vec{e}_\phi + \frac{\partial u}{\partial z} \vec{e}_z$$

$$\nabla \cdot \vec{A} = \frac{1}{\rho} \frac{\partial(\rho A_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \vec{A} = \frac{1}{\rho} \begin{vmatrix} \vec{e}_\rho & \rho \vec{e}_\phi & \vec{e}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left( \rho \frac{\partial u}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 u}{\partial \phi^2} + \frac{\partial^2 u}{\partial z^2}$$

## 5. 球坐标系 $(r, \theta, \phi)$

$$\nabla u = \frac{\partial u}{\partial r} \vec{e}_r + \frac{1}{r} \frac{\partial u}{\partial \theta} \vec{e}_\theta + \frac{1}{r \sin \theta} \frac{\partial u}{\partial \phi} \vec{e}_\phi$$

$$\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial(r^2 A_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(\sin \theta A_\theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \vec{e}_r & r \vec{e}_\theta & r \sin \theta \vec{e}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

$$\nabla^2 u = \frac{1}{r^2} \frac{\partial}{\partial r} \left( r^2 \frac{\partial u}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial u}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \phi^2}$$

梯度: 长度直接偏导.

角度乘系数, 系数只与前有关

散度: 长度整体不变, 但首次  $\times 1/\rho$  或  $r^2$   
角度乘系数.

旋度. 提  $\frac{1}{\rho}$  角度乘  $\rho$

$$\frac{1}{r^2 \sin \theta}$$

$$r, r \sin \theta$$

作业 1-2 1-4 1-7 1-9

1-14 1-23 1-24



## 第二章 静电场

### 1. 库仑定律与电场强度

库仑定律  $\vec{F} = \frac{q_1 q_2 \vec{R}}{4\pi\epsilon_0 R^3}$

场强  $\vec{E} = \frac{q\vec{R}}{4\pi\epsilon_0 R^2}$   $\epsilon_0 = \frac{1}{36\pi} \times 10^{-9} \text{ F/m}$

高斯定理  $\oint \vec{E} \cdot d\vec{s} = \frac{\sum q}{\epsilon_0} = \frac{\int \rho dv}{\epsilon_0}$

### 2. 电位

$\varphi(\vec{r}) = \frac{q}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|} = \int_P^{\vec{r}'} \vec{E} \cdot d\vec{l}'$

$E = -\nabla\varphi$   $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$   $\nabla \cdot \nabla\varphi = -\frac{\rho}{\epsilon_0}$

### 3. 电偶极子

极矩:  $\vec{p} = q\vec{l}$

### 4. 电介质中的场方程

极化强度  $\vec{P} = \lim_{\Delta V \rightarrow 0} \frac{\sum \vec{p}}{\Delta V}$   $\vec{P} = \epsilon_0 \chi_e \vec{E}$

束缚电荷 体:  $\rho(\vec{r}) = -\nabla \cdot \vec{P}(\vec{r})$

面:  $\rho_{sp} = \vec{P}(\vec{r}) \cdot \vec{n}$

真空中场方程  $\nabla \times \vec{E} = 0$   
 $\nabla \cdot \vec{E} = \frac{\rho}{\epsilon_0}$

介质中场方程  $\nabla \times \vec{E} = 0$   
 $\nabla \cdot \vec{D} = \rho$

积分  $\oint_L \vec{E} \cdot d\vec{l} = 0$   
 $\oint_V \vec{D} \cdot d\vec{s} = Q$

介电常数:  $\vec{D} = \epsilon_0(1 + \chi_e)\vec{E} = \epsilon_0 \epsilon_r \vec{E} = \epsilon \vec{E}$

### 5. 静电场的边界条件 (场量在介质交界面上的变化规律)

法向  $D_{2n} - D_{1n} = \rho_s$

切向  $E_{1t} = E_{2t}$

$\frac{\epsilon_1}{\epsilon_2} \uparrow \vec{n}$

电位  $\begin{cases} \epsilon_1 \frac{\partial \varphi_1}{\partial n} - \epsilon_2 \frac{\partial \varphi_2}{\partial n} = \rho_s \\ \varphi_1 = \varphi_2 \end{cases}$

### 6. 导体系统的电容

$\varphi_i = \sum_{j=1}^n P_{ij} q_j$   $q_i = \sum_{j=1}^n \beta_{ij} \varphi_j$

$\left. \begin{aligned} q_1 &= C_{11}\varphi_1 + C_{12}(\varphi_1 - \varphi_2) + \dots + C_{1n}(\varphi_1 - \varphi_n) \\ q_2 &= C_{21}(\varphi_2 - \varphi_1) + C_{22}\varphi_2 + \dots + C_{2n}(\varphi_2 - \varphi_n) \end{aligned} \right\}$

### 7. 电场能量

$W_e = \frac{1}{2} \sum_{i=1}^n q_i \varphi_i = \frac{1}{2} \int_V \vec{E} \cdot \vec{D} dv = \frac{1}{2} \int_V \vec{E}^2 dv$

能量密度  $w_e = \frac{1}{2} \vec{E} \cdot \vec{D}$

$W = \frac{1}{2} \epsilon_0 \int_a^b E^2 dv$

$= \frac{1}{2} QU = \frac{1}{2} CU^2$



# 第3章 恒定电流场与恒定磁场

## 1. 恒定电场

$$\vec{J} = \lim_{\Delta S \rightarrow 0} \frac{\Delta I}{\Delta S} \vec{e}_n$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t} \quad (\text{恒定电流场 } \nabla \cdot \vec{J} = 0)$$

$$\oint_S \vec{J} \cdot d\vec{s} = -\frac{dq}{dt} \quad \nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\vec{J} = \sigma \vec{E} \rightarrow \text{电导率} \quad \nabla \times \vec{J} = 0$$

$$\text{边界条件} \quad \text{积分形式} \quad \oint \vec{J} \cdot d\vec{s} = 0 \quad \oint \vec{E} \cdot d\vec{l} = 0$$

$$\text{微分形式} \quad \nabla \cdot \vec{J} = 0 \quad \nabla \times \vec{E} = 0 \quad J_{in} = J_{out} \quad E_t = E_t \quad \text{电位 } \varphi_1 = \varphi_2 \quad G \frac{\partial \varphi_1}{\partial n} = G \frac{\partial \varphi_2}{\partial n}$$

$$\text{功率损耗 } P_1 = \vec{E} \cdot \vec{J} \quad \text{同向 } P = EJ = \sigma E^2 = \frac{J^2}{\sigma} \quad (\text{单位体积})$$

$$\text{能量 } W = \frac{1}{2} \epsilon E^2$$

恒定电流场与静电场的比拟

|      |                                                            |
|------|------------------------------------------------------------|
|      | 恒定电场 (电源外)                                                 |
| 基本方程 | $\nabla \times \vec{E} = 0 \quad \nabla \cdot \vec{J} = 0$ |
| 本构关系 | $\vec{J} = \sigma \vec{E}$                                 |
| 边界条件 | $J_{in} = J_{out} \quad E_t = E_t$                         |
| 积分关系 | $I = \int \vec{J} \cdot d\vec{s}$                          |

|                                                            |
|------------------------------------------------------------|
| 静电场 ( $\rho = 0$ )                                         |
| $\nabla \times \vec{E} = 0 \quad \nabla \cdot \vec{D} = 0$ |
| $\vec{D} = \epsilon \vec{E}$                               |
| $D_{in} = D_{out} \quad E_t = E_t$                         |
| $q = \int \vec{D} \cdot d\vec{s}$                          |

例已知  $C = \frac{2\pi\epsilon}{\ln \frac{b}{a}}$  则  $G = \frac{2\pi\sigma}{\ln \frac{b}{a}}$

|                        |
|------------------------|
| 已知电荷分布                 |
| $\vec{J} \sim \vec{D}$ |
| $\epsilon \sim 0$      |
| $q \sim I$             |
| $\varphi \sim \varphi$ |

## 2. 磁感应强度、磁通及磁感线

$$\vec{F} = q\vec{v} \times \vec{B} = I d\vec{l} \times \vec{B}$$

$$\text{磁矩 } \vec{m} = I\vec{S} \quad \text{力矩 } \vec{T} = I(\vec{S} \times \vec{B}) = \vec{m} \times \vec{B} = \vec{r} \times \vec{F}$$

$$\text{磁通 } \Phi = \int_S \vec{B} \cdot d\vec{s}$$

## 3. 真空中恒定磁场

$$\oint_L \vec{B} \cdot d\vec{l} = \mu_0 I$$

$$\oint_S \vec{B} \cdot d\vec{s} = 0$$

$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

$$\nabla \cdot \vec{B} = 0$$

毕奥-萨伐尔定律

$$\vec{B} = \frac{\mu_0}{4\pi} \oint_L \frac{I d\vec{l} \times \vec{R}}{R^3}$$

## 4. 矢量磁位与标量磁位

$$\vec{B} = \nabla \times \vec{A} \quad \nabla \cdot \vec{A} = 0$$

$$\nabla^2 \vec{A} = -\nabla \times \vec{B} = -\mu_0 \vec{J}$$

$$\Phi = \int_S \vec{B} \cdot d\vec{s} = \int_S (\nabla \times \vec{A}) \cdot d\vec{s} = \oint_L \vec{A} \cdot d\vec{l}$$

## 5. 磁介质

$$\text{磁化强度 } \vec{M} = \sum_{i=1}^N \vec{m}_i$$

$$\vec{A}(\vec{r}) = \frac{\mu_0}{4\pi} \frac{\vec{M} \times (\vec{r} - \vec{r}')}{|\vec{r} - \vec{r}'|^3}$$

$$\text{磁化电流密度 } \vec{J}' = \nabla \times \vec{M}$$

$$\vec{J}' = \nabla \vec{M} \times \vec{e}_n$$

边界条件

恒定电流场

$$\nabla \cdot \vec{J}' = 0 \quad J'_{in} = J'_{out}$$

$$\nabla \times \vec{E}' = 0 \quad E'_{t1} = E'_{t2}$$

静电场

$$\nabla \cdot \vec{D}' = 0 \quad D'_{n1} = D'_{n2}$$

$$\nabla \times \vec{E}' = 0 \quad E'_{t1} = E'_{t2}$$

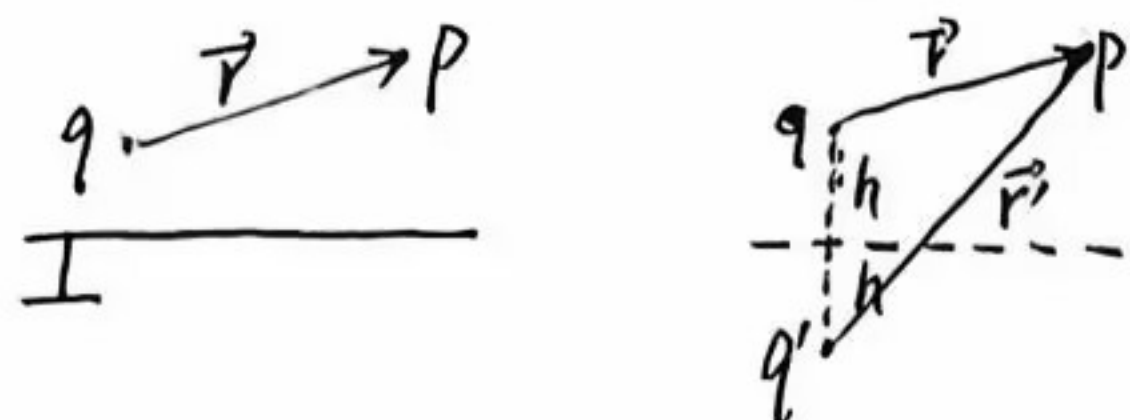
$$\mu_0 = 4\pi \times 10^{-7}$$



# 第4章 静电场的边值问题

## 1. 镜像法

点电荷与无限大接地平的导体平面



引入镜像电荷的原则

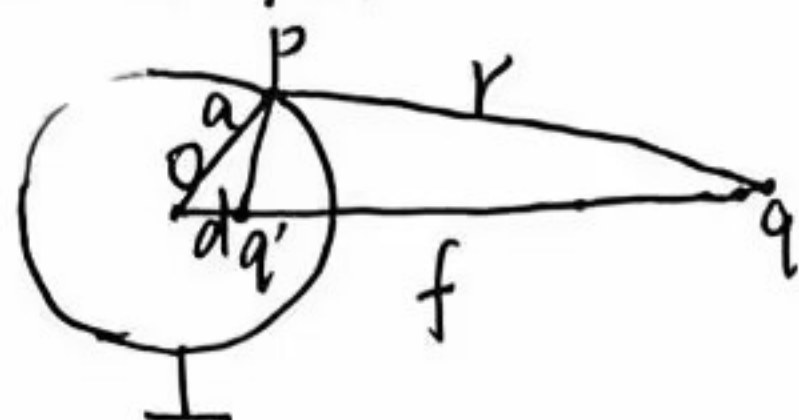
- ① 位于求解区域之外
- ② 前后边界条件不变

求解  $q'$ . 利用  $\varphi|_{z=0} = 0$

$$q' = -q$$

$$\varphi = \frac{q}{4\pi\epsilon_0 h} + \frac{q'}{4\pi\epsilon_0 h} = 0$$

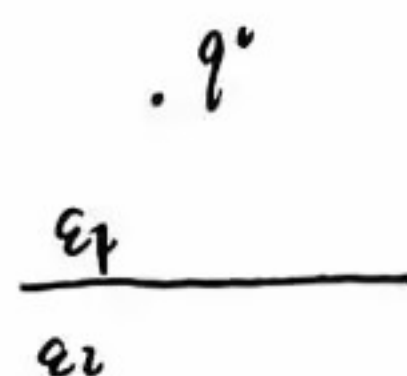
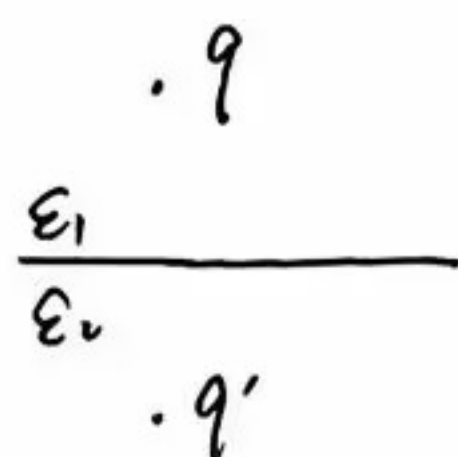
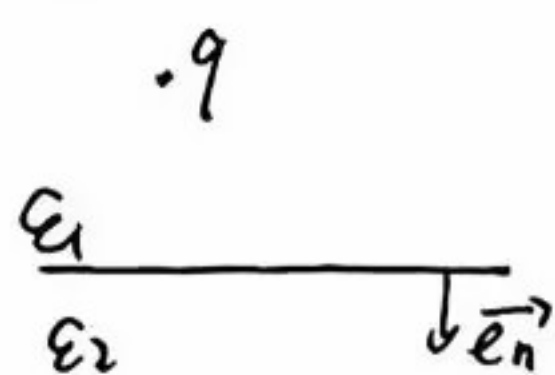
点电荷与导体球



$$q' = -\frac{a}{f} q$$

$$d = \frac{a^2}{f}$$

点电荷与无限大介质平面



$$q' = \frac{\epsilon_1 - \epsilon_2}{\epsilon_1 + \epsilon_2} q$$

$$q'' = \frac{2\epsilon_2}{\epsilon_1 + \epsilon_2} q$$

## 2. 分离变量法

$\varphi(0, y) = \varphi(a, y) = \text{常数}$ .  $\rightarrow$  在  $x$  方向选三角函数

$= 0$  选正弦  $\neq 0$  选余弦

$$k_x^2 + k_y^2 = 0$$

$\rightarrow$  在  $y$  方向上选双曲函数或指数函数.

$\downarrow$   
有限. 无限



## 第5章 时变电磁场

### 1. 求感应电动势

$$\mathcal{E} = -\frac{d\Phi}{dt} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S} + \oint_L (\vec{v} \times \vec{B}) \cdot d\vec{l}$$

### 2. 求位移电流

$$\vec{J}_d = \frac{\partial \vec{D}}{\partial t} = \epsilon \frac{\partial \vec{E}}{\partial t} \quad \nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

$$\text{传导电流 } \vec{J}_c = \sigma \vec{E} \quad \text{电容中 } \vec{J}_c = C \frac{dU}{dt}$$

### 3. 麦克斯韦方程组

积分  $E D B H$

$$\oint_L \vec{H} \cdot d\vec{l} = \int_S (\vec{J} + \frac{\partial \vec{D}}{\partial t}) \cdot d\vec{S}$$

$$\oint_L \vec{E} \cdot d\vec{l} = -\int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{S}$$

$$\oint_S \vec{B} \cdot d\vec{S} = 0$$

$$\oint_S \vec{D} \cdot d\vec{S} = q$$

微分

$$\nabla \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

$$\nabla \cdot \vec{B} = 0$$

$$\nabla \cdot \vec{D} = \rho_s$$

辅助方程

$$\vec{D} = \epsilon \vec{E}$$

$$\vec{B} = \mu \vec{H}$$

$$\vec{J} = \sigma \vec{E}$$

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$

### 4. 边界条件

$$\vec{n} \times (\vec{H}_2 - \vec{H}_1) = \vec{J}_s$$

$$\vec{n} \times (\vec{E}_2 - \vec{E}_1) = \vec{0}$$

$$\vec{n} \cdot (\vec{B}_2 - \vec{B}_1) = 0$$

$$\vec{n} \cdot (\vec{D}_2 - \vec{D}_1) = \rho_s$$

### 5. 坡印廷矢量

$$\text{瞬时值 } \vec{S} = \vec{E} \times \vec{H}$$

$$\text{平均值 } S_{av} = \frac{1}{2} \text{Re}(\vec{E} \times \vec{H}^*)$$

$$\text{功率 } P = \int_S (\vec{E} \times \vec{H}) \cdot d\vec{S}$$

### 6. 位函数

$$\text{时变场 } \vec{E} = -\nabla\varphi - \frac{\partial \vec{A}}{\partial t}$$

$$\vec{B} = \nabla \times \vec{A}$$

(cosθ)  
矢量瞬时值与复数值相互表示

$$\sin(\omega t - kx) \leftrightarrow -je^{-jkx}$$



# 第6章 均匀平面电磁波

## 1. 无耗媒质中的平面电磁波

$$\vec{E} = \vec{e}_x E_m \cos(\omega t - kz + \phi_0) = \vec{e}_x E_0 e^{-jkz} \quad (\text{复数值})$$

瞬时

$$\vec{H} = \vec{e}_y \frac{1}{\eta} E_m \cos(\omega t - kz + \phi_0) = \vec{e}_y \frac{1}{\eta} E_0 e^{-jkz}$$

$e^{-jkz}$  向+z方向传播  
 $e^{jkz}$  -z

$$\eta = \sqrt{\frac{\mu}{\epsilon}} \quad \text{媒质本征波阻抗}$$

$$\eta_0 = \sqrt{\frac{\mu_0}{\epsilon_0}} = 120\pi \approx 377 \Omega$$

$$\text{周期 } T = \frac{2\pi}{\omega} \quad \text{频率 } f = \frac{1}{T} = \frac{\omega}{2\pi}$$

$$\text{波长相 } \lambda = \frac{2\pi}{k} \quad \text{相移常数 } k = \frac{2\pi}{\lambda} = \omega \sqrt{\mu\epsilon} \quad \text{相速度 } v = \frac{\omega}{k} = \frac{1}{\sqrt{\mu\epsilon}} = \frac{c}{f} \rightarrow \text{自由空间}$$

在理想介质中, 均匀平面波的电场强度与磁场强度相互垂直且同相位

$$\vec{E} = \eta \vec{H} \times \hat{k} \quad \vec{E}, \vec{H}, \hat{k} \text{ 三者垂直}$$

$$\vec{H} = \frac{1}{\eta} \hat{k} \times \vec{E}$$

$$\text{电场能量密度 } W_e = \frac{1}{2} \epsilon E^2 = \frac{1}{2} \mu H^2 = W_m \quad \text{磁场能量密度}$$

$$\text{电磁波能量密度 } W = W_e + W_m = \epsilon E^2 = \mu H^2$$

$$\text{电磁波能流密度 } \vec{S} = \vec{E} \times \vec{H} = \vec{E} \times \frac{1}{\eta} \hat{k} \times \vec{E} = \frac{1}{\eta} E^2 \hat{k}$$

$$S_{av} = \frac{1}{2} \text{Re} [\vec{E} \times \vec{H}^*] = \frac{1}{2\eta} E_0^2 \hat{k}$$

## 2. 媒质导电性对场的影响

根据  $\frac{\sigma}{\omega\epsilon}$  衰减常数

|             |     |
|-------------|-----|
| $\gg 1$     | 良导体 |
| $\ll 1$     | 弱导体 |
| $\approx 1$ | 半导体 |

有损耗  $v = \frac{\omega}{\beta}$

\* 良导体 电场相位超前磁场

$$\alpha \approx \beta = \sqrt{\frac{1}{2} \omega \mu \sigma} = \sqrt{\pi f \mu \sigma} \quad \text{衰减常数 } \beta$$

$$\tilde{\eta}_c = \sqrt{\frac{j\omega\mu}{\sigma}} e^{j\frac{\pi}{4}} \quad \text{波阻抗}$$

$$\delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\omega \mu \sigma}} \quad \text{趋肤深度} = \frac{1}{2\pi}$$

$$v = \frac{\omega}{\beta}$$

$$\lambda = \frac{2\pi}{\beta}$$

弱导体

$$\alpha \approx \frac{\sigma}{2} \sqrt{\frac{\mu}{\epsilon}} \quad \beta \approx \omega \sqrt{\mu\epsilon} \quad \tilde{\eta} \approx \sqrt{\frac{\mu}{\epsilon}}$$

$$\mu_0 = 4\pi \times 10^{-7}$$

## 3. 电磁波的极化

线极化波  $\phi_1 - \phi_2 = 0$  或  $\pi$

圆极化波  $E_{m1} = E_{m2}$ ,  $\phi_1 - \phi_2 = \frac{\pi}{2}$  右旋 沿z轴正向传播, 反向反之

$\phi_1 - \phi_2 = -\frac{\pi}{2}$  左旋

椭圆极化波  $\pi > \phi_1 - \phi_2 > 0$  右旋

$-\pi < \phi_1 - \phi_2 < 0$  左旋

① 先看相位, 把“-j”

“sin”化成标准

② 看传播方向

左, 右旋



# 第7章 电磁波的反射和折射

## 1. 对理想导体的分界面的垂直入射

$z < 0$  理想介质  $\mu_1, \epsilon_1, \sigma_1 = 0$

感应电流  $\vec{J}_s = \vec{n} \times \vec{H}_s$

$z > 0$  理想导体  $\sigma_2 \rightarrow \infty$

入射波电场  $\vec{E}^+ = \vec{e}_x E_m^+ e^{-jkz}$  磁场  $\vec{H}^+ = \frac{1}{\eta} \vec{e}_y E_m^+ e^{-jkz}$

反射波电场  $\vec{E}^- = -\vec{e}_x E_m^+ e^{jkz}$  磁场  $\vec{H}^- = \frac{1}{\eta} \vec{e}_y E_m^+ e^{jkz}$

$z < 0$  合成场  $\vec{E}_s = \vec{E}^+ + \vec{E}^- = \vec{e}_x E_m^+ (e^{-jkz} - e^{jkz}) = -j\vec{e}_x 2E_m^+ \sin kz$

$\vec{H}_s = \vec{H}^+ + \vec{H}^- = \vec{e}_y \frac{E_m^+}{\eta} (e^{-jkz} + e^{jkz}) = \vec{e}_y \frac{2}{\eta} E_m^+ \cos kz$

## 2. 对两种理想介质分界面的垂直入射

$z < 0$   $\mu_1, \epsilon_1$   $z > 0$   $\mu_2, \epsilon_2$

反射系数  $\Gamma = \frac{E_{rm}}{E_{im}} = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$

透射系数  $T = \frac{E_{tm}}{E_{im}} = \frac{2\eta_2}{\eta_2 + \eta_1}$

$\Gamma = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1}$

光密  $\rightarrow$  光疏  $\epsilon_1 > \epsilon_2$   $\eta_1 < \eta_2$   
 $T > 0$

$|\vec{E}|$  最大  $2kz = -2n\pi$  ( $n=0, 1, 2, \dots$ )

$z = -\frac{n\lambda}{2} = -\frac{n\lambda}{2}$   $E_{smax} = E_{im}(1+T)$

$|\vec{E}|$  最小  $2kz = -(2n+1)\pi$

$z = -\frac{(2n+1)\lambda}{4}$   $E_{smin} = E_{im}(1-T)$

功率密度 入射波  $S_{iav} = \frac{1}{2} \text{Re}[\vec{E}_i \times \vec{H}_i^*] = \vec{e}_z \frac{1}{2\eta_1} E_{im}^2$

反射波  $S_{rav} = \frac{1}{2} \text{Re}[\vec{E}_r \times \vec{H}_r^*] = -\vec{e}_z \frac{1}{2\eta_1} \Gamma^2 E_{im}^2$

透射波  $S_{tav} = \frac{1}{2} \text{Re}[\vec{E}_t \times \vec{H}_t^*] = \vec{e}_z \frac{1}{2\eta_2} T^2 E_{im}^2$

## 3. 对一般导体媒质分界面的垂直入射

复数  $\eta \rightarrow \eta_c = \sqrt{\frac{\mu}{\epsilon - j\sigma}}$   $k$

## 4. 均匀平面波对理想介质分界面的斜入射

Snell定理  $k_1 \sin \theta_i = k_1 \sin \theta_r = k_2 \sin \theta_t \Rightarrow \begin{cases} \theta_r = \theta_i \\ \frac{\sin \theta_i}{\sin \theta_t} = \frac{k_2}{k_1} = \frac{\omega \sqrt{\epsilon_2 \mu_1}}{\omega \sqrt{\epsilon_1 \mu_1}} \end{cases}$

临界角: 刚好发生全反射  $\theta_c = \arcsin \sqrt{\frac{\epsilon_2}{\epsilon_1}}$

媒质2为理想导体  $T=0$   $\Gamma=-1$   
电磁波垂直入射到理想导体面上反射系数  
反射与入射相位差相差180° 一半波损失

驻波系数  $S = \frac{|E|_{max}}{|E|_{min}} = \frac{1+|\Gamma|}{1-|\Gamma|} \Rightarrow |\Gamma| = \frac{S-1}{S+1}$

$T=0, S=1$  行波

$T=\pm 1, S \rightarrow \infty$  纯驻波

$0 < |\Gamma| < 1, 1 < S < \infty$  混合波

$S$  越大, 驻波越大, 行波越小